Supplementary Material
Native Association: Confidence-Aware Human Perception in the Wild with a Foundation VLM

Igal Dmitriev and Ofir Liba

WSC Sports
{igal.dmitriev,ofir.liba}@wsc-sports.com

1 Extended Figures

This section collects conceptual and diagnostic figures deferred from the main paper for space. Figure S2 illustrates the abstention mechanism behind the frontier APIs' apparent precision edge, and Tab. S1 gives the conditional (given-detected) jersey-recall breakdown behind it; Fig. S3 summarizes the single recipe applied to both domains; Fig. S4 explains why small-text reading is resolution- rather than knowledge-bounded; Fig. S5 is the confidence ROC and distribution referenced from §5.4 of the main paper, and Fig. S6 illustrates the per-jersey confidence gate deferred from §3.2; Fig. S7 is the full vector schematic of the crop re-query. Figure S8 is the size-stratified plot referenced from §5.2 of the main paper (the plot form of Tab. S3), and Fig. S11 is the risk–coverage curve referenced from §5.4 of the main paper, with Tab. S6 tabulating its operating points and Tab. S8 the matched-coverage comparison behind §5.1; Tab. S7 is the confidence-aggregation ablation referenced from §3.2 of the main paper (geometric vs. arithmetic mean vs. min-probability over the same replay logits), and Fig. S12 is the qualitative comparison on a crowded frame referenced from §5.1. Figure S1 visualizes the single grammar-constrained sequence described by the serialization block in §3.1 of the main paper. Figures S9 and S10 show qualitative WIDER-Attribute outputs.

Team-head confidence. The confidence story extends beyond jersey reads to the team head. From the stored per-player `team_probs` we take the classification margin (top-1 minus top-2 probability) and ask whether it predicts team correctness on the IoU-matched players (correctness defined after the optimal per-frame A↔B alignment, the same convention as team purity, since the cluster labels are arbitrary). The margin is an informative signal—AUROC 0.835 at team accuracy 0.970 (Tab. S2)—so the same rank-and-abstain reject option that works for jersey reads also applies to team assignment. The crop re-query (E3) leaves the team head untouched, so its numbers are identical to E1b. Per-item OCR-*text* confidence was not dumped in this run; *jersey_conf* covers the primary digit item only.

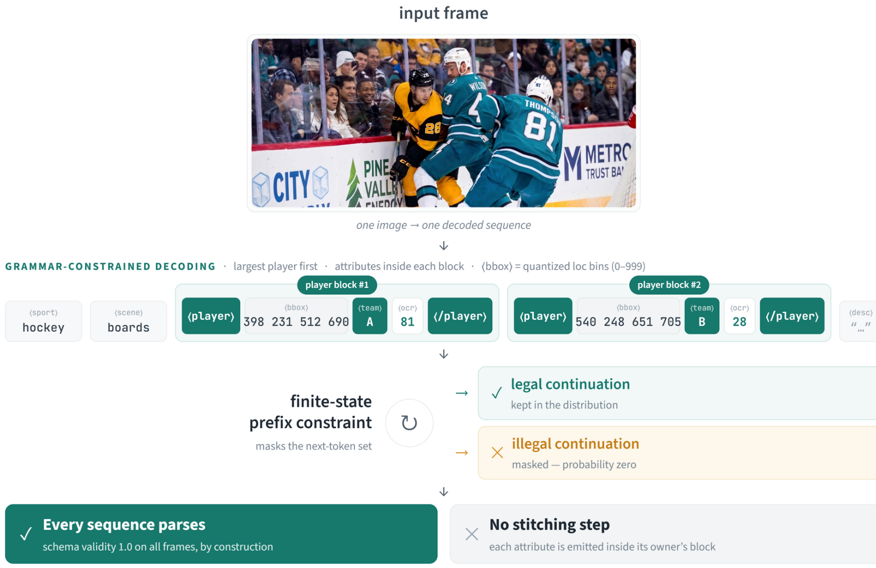

Fig. S1: One image, one grammar-constrained sequence—the visual form of the serialization in §3.1 of the main paper. Structural tokens open a block per person (largest first; boxes emitted as quantized location bins), and at each step a finite-state prefix constraint masks illegal continuations to probability zero. Every decoded sequence therefore parses (schema validity 1.0 on all frames, by construction) and no post-hoc stitching step exists—each attribute is emitted inside the block of the person it belongs to.

WIDER-Attribute phenomenology (mechanism). The three transferred findings of §6 of the main paper behave mechanistically as on sports (Fig. S3). The confidence signals split the same way: the per-attribute probabilities, fused across attributes, are calibrated (ECE 0.028) and carry the reject option (4.1% → 1.5% error at 10% abstention), while the structural signals are uninformative—the localization/objectness confidence does not separate correct from incorrect decisions, so it routes the re-query rather than filtering false positives, exactly the negative result reported on sports (§5.4). The tuning null is likewise a mirror: a stronger-tuned checkpoint differs from the base by +0.02 mAP given-box and −0.8 mA, both inside noise. The gains, on both domains, live in the output structure rather than in additional weights.

The uncertainty token doubles as a visibility detector. The fixed-slot `<unspecified>` token also learns real signal: its probability detects annotator-unspecified slots with AUC 0.82–0.94 (strongest on lower-body attributes occluded in crowds)—the schema's uncertainty slot doubles as a visibility detector at no extra cost.

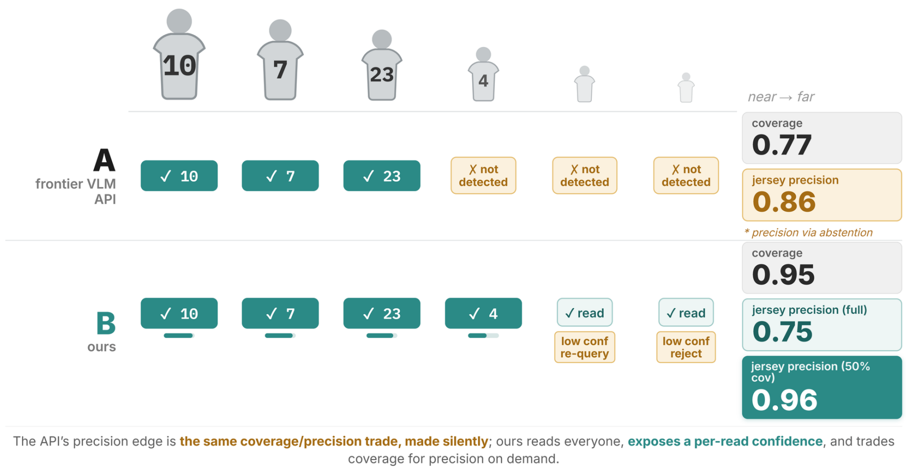

Fig. S2: Precision *via* abstention. A frontier VLM API (Gemini-3.1) posts high jersey precision (0.86) only by silently skipping the hard, small players (coverage 0.77); our model reads everyone (coverage 0.95) at precision 0.75, exposes a per-read confidence, and can therefore trade coverage for precision on demand (0.96 when gated to 50%). The API's "precision edge" is the same coverage/precision trade, made silently rather than explicitly (operating points from Table 1 of the main paper; cf. the reject option, main paper §5.4). Per-player marks are illustrative; the chips are measured test-set aggregates.

2 Cascade Performance by Player Size

For completeness we give the size-stratified reading quality of the two oracle-boxed cascades (Tab. S9), omitted from the main paper's size table to avoid a twelve-column single-column layout. As with the aggregate numbers, detection recall is *GT* for all buckets (the cascades are granted ground-truth boxes), so these cells report reading quality on the `full_frame`+stitch oracle tier only. The story mirrors the main paper: cascade jersey-F1 degrades on the small and tiny buckets even with perfect boxes, because their reading—not their localization—is the bottleneck.

3 When Association Still Fails

Two residual failure modes remain even for our native-association model, and both trace to the same cause—overlap and illegibility, not a broken binding step. We first define the association metric and decompose it (Sec. 3.1), then quantify *crop-bleed* on the oracle cascades (the clean control), then show the analogous residual failures on our own outputs (Fig. S13).

Table S1: Conditional jersey recall (the honesty view behind the abstention story, referenced from §5.1 of the main paper). *Unconditional* recall is over all ground-truth players with a jersey number (matches the main-paper jersey R); *given-detected* restricts the denominator to IoU-matched players; *coverage* is the matched fraction. Conditioned on detection the APIs read as well as or better than we do—their lower unconditional recall is a *detection* deficit (low coverage), not a reading one. Same matcher and GT vintage as Tab. S10.

System	Uncond. recall	Given-detected	Coverage
Ours (E1b)	0.759	0.783	**0.950**
Ours +re-query (E3)	0.799	0.824	**0.950**
Gemini-2.5-pro	0.612	**0.834**	0.704
Gemini-3.1-pro	0.645	0.821	0.772

Table S2: Team-head margin as a correctness predictor on the frozen test (matched players; margin = top-1−top-2 of `team_probs`). Correctness uses the optimal per-frame A↔B alignment. E3 is identical to E1b (re-query does not touch the team head). Generated by `team_margin_auroc.py`. Aligned accuracy applies the per-frame optimal A↔B correspondence (the purity convention).

System	Matched players	Aligned acc.	AUROC (margin)
Ours (E1b)	1203	0.970	0.835
Ours +re-query (E3)	1203	0.970	0.835

3.1 Jersey-digit AER: definition and decomposition

The headline association error rate is computed over correctly-read jersey *digits* only. Every correctly-read OCR token bound to the wrong player falls into exactly one of three disjoint classes, so the all-text rate reconciles as all-misassoc = digit-swap + digit-unmatched + team-bleed (Tab. S10): a **digit-swap** is a jersey read on a prediction that IoU-matched the *wrong* real player (a genuine identity swap); a **digit-unmatched** read is a jersey read grounded to no valid box (an unmatched or degenerate detection)—a number read correctly but attached to no athlete; **team-bleed** is a non-digit read (team/league text) on a teammate, whose owner is ill-defined because the text is identical across teammates. The headline digit-AER = (digit-swap + digit-unmatched)/(correctly-read jersey digits); team-bleed is excluded. We also report the *matched-only* rate (swap / digits) for transparency, but it is *not* the headline: it counts only failures that landed on a real box, so it rewards a system for grounding a read to nothing rather than to the wrong player—the same abstention/coverage trap as jersey precision. On this metric the APIs would appear near-perfect (0.007–0.023) precisely because 90–97% of their misbound digits are grounded to no box (Tab. S10, digit-unmatched row); that is the failure the metric must count, so we do not use it as the headline.

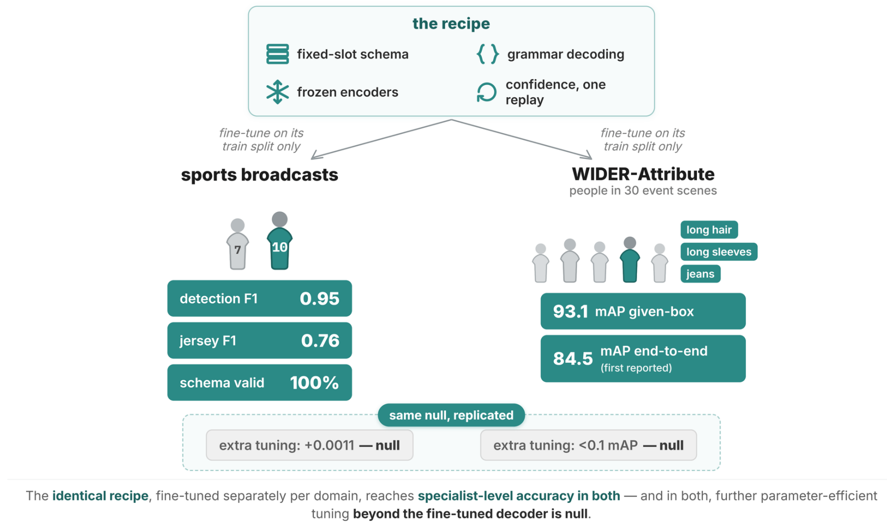

Fig. S3: One recipe, two worlds. The identical recipe (fixed-slot schema, grammar-constrained decoding, frozen encoders, confidence from one replay), fine-tuned separately on each domain's own train split, reaches specialist-level accuracy on both sports broadcasts and WIDER-Attribute. In both, further parameter-efficient tuning *beyond the fine-tuned decoder* is a null (sports: at most +0.0011 on a 0–1 scale; WIDER: < 0.1 mAP).

Stitch-dropped reads (cascades). The oracle cascades read the full frame and bind each OCR read to a player only when its IoSmaller with some ground-truth box clears 0.5; a read that clears no box is *dropped*, which would erase it from both the numerator and the denominator—the cascade's exact analogue of the APIs' grounded-to-nothing digit, which we insisted on counting. We therefore recover, from the pre-stitch `ocr_items_raw`, every correct-digit read whose quad overlaps a player body yet the stitch dropped (the on-body clause excludes scoreboard/crowd digits) and add it to both. The measured count is 0 for both cascades: with oracle *full-body* boxes an on-body jersey-digit quad is substantially contained (IoSmaller ≥ 0.5) and always binds, so only off-body text is dropped (2 reads for `florence_ff`, 1 for `paddle_ff`, all scoreboard/ crowd). The cascade digit-AER is thus matched-swaps-only *as measured*, not by construction, and the near-tie against E1b (0.057) survives this counting.

History. The metric choice is disclosed in §4 of the main paper; operationally: earlier drafts scored the all-text rate (E1b 0.085), which is dominated ($\approx$ 76% of our misassociations) by team-name bleed—text identical across teammates, hence not an identity error. The all-text row is retained in Tab. S10 for every system; note the oracle cascades' lower all-text values (0.057–0.070) are coverage-confounded (they emit roughly a third of the reads, and team-bleed opportunity scales with reading coverage)—one illustration of why all-text is not a valid

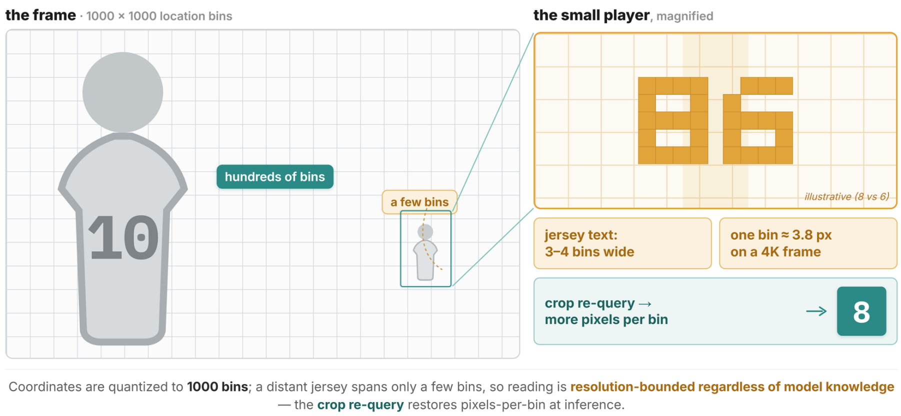

Fig. S4: Quantization, not knowledge, bounds small-text reading. Coordinates are quantized to 1000 location bins, so on a 4K frame one bin is ≈ 3.8 px and a distant jersey spans only 3–4 bins—reading is resolution-bounded regardless of model knowledge, and the crop re-query restores pixels-per-bin at inference time. The exact digits in the magnified panels are illustrative only (the "86"/"8" tiles depict the low-resolution ambiguity—e.g. "8" vs. "6"—not a specific ground-truth number).

identity metric. Our own duplicate- and degenerate-box reads remain charged (in digit-unmatched); the switch tightens the metric without special-casing our failures.

3.2 Per-crop crop-bleed

The main paper reports that per-crop cascades, granted ground-truth boxes and reading each player from its *own* crop, still misattribute 15–18% of jersey digits. Here we document how that number is obtained and why it is the cleanest available evidence that association is not solved by localization.

Table S3: Size-stratified performance on the frozen test (GT-box-area quartiles; ≈ 317 matched players per bucket), referenced from §5.2 of the main paper. Detection recall (R_{det}), jersey F1, and OCR-text F1 per bucket; *+req* is the training-free crop re-query (E3). Detection recall is unchanged by re-query (same detector). Gemini-3.1-pro is the medium-resolution configuration. Cascade rows are in Tab. S9.

	R_{det}		Jersey F1			OCR-text F1	
Bucket	Ours	Gem-3.1	Ours	+req	Gem-3.1	Ours	+req
tiny	0.849	0.558	0.606	**0.746**	0.532	0.497	**0.616**
small	0.975	0.763	0.741	0.753	0.726	0.552	0.559
med	0.987	0.873	0.822	0.818	0.828	0.592	0.597
large	0.991	0.896	0.832	0.839	0.790	0.656	0.660

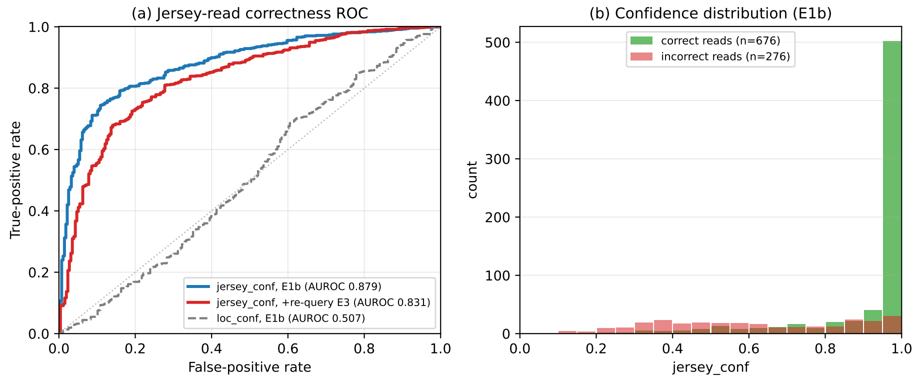

Fig. S5: Per-field confidence on the frozen test (referenced from §5.4 of the main paper). **(a)** ROC of *jersey_conf* as a correctness predictor: AUROC 0.879 (E1b) and 0.831 (+re-query, E3); the structural *loc_conf* (dashed) is near-random (0.507), a negative result we report explicitly. **(b)** Distribution of *jersey_conf* over correct vs. incorrect reads (E1b): correct reads pile up near 1 while incorrect reads spread lower, so ranking-and-abstaining trades coverage for precision.

Table S4: LoRA-target ladder vs. the no-adapter base (aggregate *val_score* on the frozen test, `best_model_score`), referenced from §5.3 of the main paper. All gains are within noise ($< +0.005$), roughly 5× under the pre-registered floor despite up to 3.15M trainable parameters—the null that motivates "structure over tuning".

Variant	r/α	Trainable params	Aggregate	Δ vs. base
Base (no adapter)	–	–	0.7831	–
+LoRA (decoder V)	16/32	0.79M	0.7832	+0.0001
+LoRA (vision)	8/16	1.08M	0.7836	+0.0005
+LoRA (decoder QVO)	16/32	2.36M	0.7835	+0.0004
+LoRA (decoder QKVO)	16/32	3.15M	0.7842	+0.0011

Order-aware source re-assignment. The published run assigns per-crop reads to boxes by an IoSmaller stitching heuristic; under occlusion this can credit a bled read to the neighbouring (often correctly-numbered) large player by luck. To charge each read to its *true* source crop we exploit the fact that `ocr_items_raw` is emitted in crop-append order (verified 100% monotone on this run): walking the list with a non-decreasing pointer, each read is bound to the earliest ground-truth box, at or beyond the pointer, that geometrically contains it. This is a purely offline recomputation—no GPU, no re-inference—and a `check` mode reproduces the published numbers byte-for-byte before the fix is applied.

Result. After re-assignment, per-crop jersey-digit association error settles at 0.151 (Florence-2) and 0.179 (PaddleOCR)—*not* near zero (all-text rates 0.135/0.145)—at read coverage (recall) 0.45–0.46, versus our 0.057 at recall 0.76.

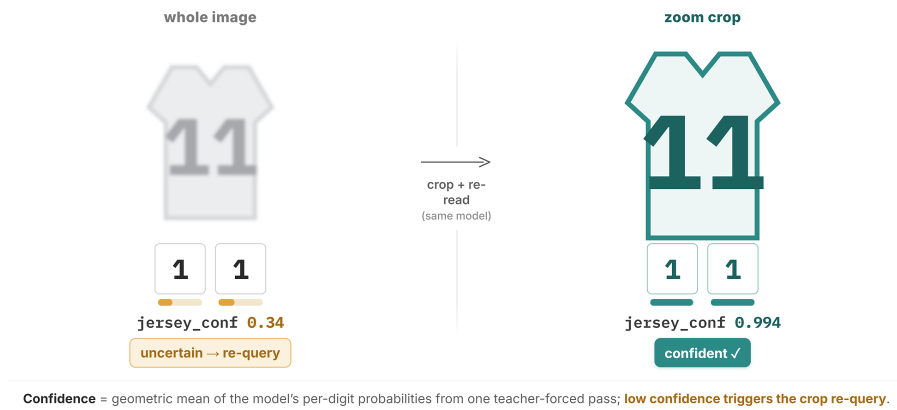

Fig. S6: Per-jersey confidence gates the read (deferred from §3.2 of the main paper). *jersey_conf* (Eq. (1) of the main paper) is the geometric-mean token probability from one teacher-forced replay; a blurred distant jersey yields 0.34 (uncertain → re-query), the zoomed crop re-read yields 0.994. The confidence values shown are illustrative, not measurements; measured distributions are in Fig. S5.

Table S5: Quad supervision on/off (both no LoRA, `best_model_score`; jersey F1 under the strict exact-digit rule). Dropping quads (E1b) is our headline configuration.

Config	Det F1	AP@50	Team pur.	Jersey F1	OCR-text F1	Lat. (ms)
Quads on (E1)	**0.954**	**0.886**	0.915	0.665	0.568	1710
Quads off (E1b)	0.950	**0.886**	**0.952**	**0.695**	**0.584**	**1224**

The cascade is thus favoured on *both* axes (oracle crops and fewer, easier reads) and still loses: overlapping athletes physically leak neighbours' numbers into the crop (*crop-bleed*). The same re-attribution moves per-crop jersey *precision* down in lock-step—0.810 → 0.787 (Florence-2), 0.798 → 0.772 (PaddleOCR)—the signature of honest attribution: reads that previously landed on the right big player by stitch-luck are now correctly charged to the wrong source crop. The precision drop and the AER are two views of the same phenomenon. Localization does not dissolve association; it must be modeled.

3.3 Residual failures on our own output

Figure S13 shows the two residual failure modes on our own predictions. The first is the perception-level association error behind the main-paper AER: at very heavy occlusion (GT pairwise overlap ≈ 0.98) a correctly-read number can still be emitted in a neighbour's block. The second is illegibility: on the tiniest jerseys the digits are under the resolution floor (Fig. S4), and even after the crop re-query the reads are truncated (e.g. "12"→"7", "15"→"5"). These are honest boundary

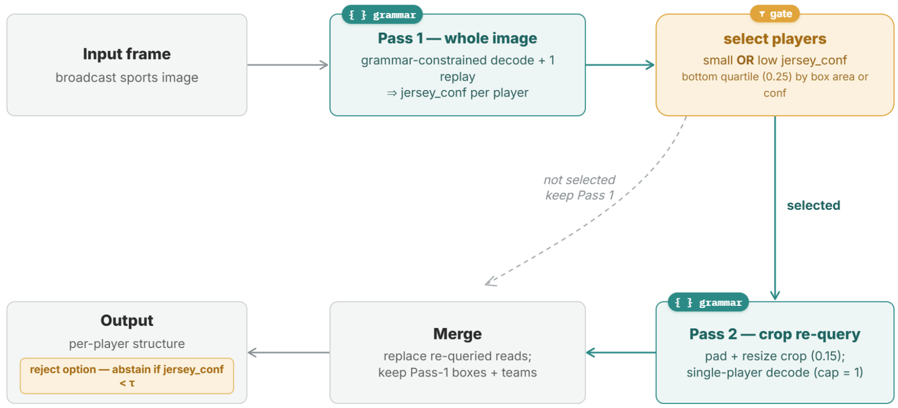

Fig. S7: Full schematic of the training-free, confidence-gated crop re-query (the 2×3 layout). Pass 1 decodes the whole frame under the grammar and attaches a *jersey_conf* to each player via one teacher-forced replay; a GT-free gate selects small or low-confidence players for a single-player Pass 2 on a padded crop; only those players' reads are replaced, and every output player is tagged with its source. Gate = bottom quartile (fraction 0.25) by predicted box area or by *jersey_conf*; Pass 2 runs the same grammar with the player cap set to one; padding fraction 0.15, validation-selected.

cases at the edges of occlusion and resolution, not the systematic identity swaps a coverage-limited post-hoc matcher produces.

3.4 Conditioning of residual misassociations

To answer *when* association still fails, we re-derive every misassociated *jersey-digit* read of the four end-to-end systems from the cached predictions with the same IoU matcher and fuzzy correctness test as the headline AER (Sec. 7; reconciled via the gate of Sec. 10) and, for each, measure the geometry between the emitting box and the true owner's ground-truth box (Tab. S11). The two families are qualitatively different. *Ours are contact-occlusion identity swaps*: 76% (E1b) to 68% (E3) of our misassociated digits land on a *matched* but wrong player, 63–87% physically overlapping the true owner (median IoSmaller 0.77–0.88) in frames with a median of two mutually overlapping boxes—the same physical cause as crop-bleed (Sec. 3.2). *The APIs' are grounded to nothing*: 90% (Gemini-2.5) and 97% (Gemini-3.1-pro, medium-resolution) of their misbound digits sit on a box that matches *no* ground-truth player. Gemini-2.5 at least places these near the true owner (94% overlap), but too imprecisely to match; Gemini-3.1-pro is the extreme case—93% of its misbinds are between *non*-overlapping players (median centre distance 0.55 of the frame, zero overlap, uncrowded), the degenerate zero-area-box behaviour noted in §5.5 of the main paper. Concretely,

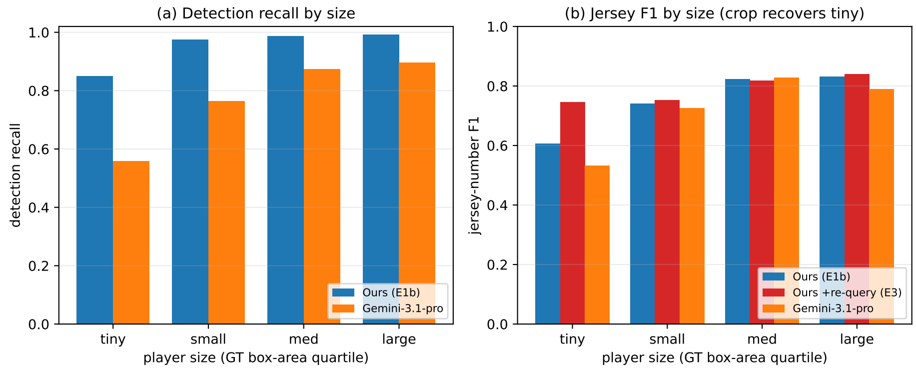

Fig. S8: Performance by player size (GT-box-area quartiles; the plot form of Tab. S3). **(a)** Detection recall: ours stays high across sizes while Gemini-3.1-pro collapses on tiny players. **(b)** Jersey F1 rises with size for all systems (reading is size-bound); the training-free crop re-query (E3) recovers the tiny bucket ($0.606 \rightarrow 0.746$) with negligible change elsewhere.

Table S6: Reject option: jersey-number precision at fixed coverage, ranking predictions by *jersey_conf* (higher coverage = fewer abstentions). Full coverage = no abstention over the scored reads (definition below). The closed APIs expose no per-item confidence; the cascades' per-read OCR scores are not confidences over the associated player–number structure.

	Coverage				
System	1.0	0.9	0.7	0.5	0.3
Ours (E1b)	0.710	0.769	0.874	0.956	0.979
Ours +re-query (E3)	0.736	0.789	0.872	0.932	0.959

our ambiguous-geometry swaps occur between ground-truth boxes overlapping at IoSmaller 0.67–0.96, while the APIs' grounded-to-nothing reads sit on boxes far from any player; rendered examples of both residual modes are in Fig. S13.

Ambiguous vs. resolvable. We further split each swap by whether the emitter's matched ground-truth box and the true owner's box overlap enough to be *geometrically unidentifiable*—one substantially inside the other (GT–GT IoSmaller > 0.5), so that no matcher could attribute the digit—versus *resolvable* (a distinguishable swap, or an unmatched orphan a correct box would place). Contrary to a natural hope, our errors are *not* concentrated in the unidentifiable regime: only 15.3% of E1b (and 37.8% of E3, which re-reads deeper into pile-ups) are unidentifiable, so the majority are honest, correctable perception errors—we do not hide them behind inherent ambiguity. The APIs are also mostly resolvable (93–98%) but for the opposite reason: their misbinds are grounded-to-nothing orphans that a valid box would fix, not geometrically

Fig. S9: WIDER-Attribute, end-to-end single-pass detection across event categories (curated examples; model predictions from the frozen WIDER checkpoint, no ground-truth boxes at any stage). One grammar-constrained pass localizes every person across in-the-wild scenes.

confusable swaps. The oracle cascades, by contrast, are the most unidentifiable of all (64–83% of their few swaps have GT–GT IoSmaller > 0.5), confirming that crop-bleed is a genuine contact-overlap phenomenon between physically inseparable boxes.

4 Jersey-Rule Sensitivity

The main-paper jersey metric credits a ground-truth number when it appears as any digit token the matched prediction emits (the *any-OCR* rule). A stricter rule—requiring the prediction's primary digit jersey to equal the ground-truth primary digit exactly—is the one used in the quad ablation of the main paper (§5.3). Table S12 re-scores every cached system under both rules with the same IoU matcher (offline, no re-inference; the guarded cascade run, Sec. 10); the strict E1/E1b values reproduce that ablation (0.665/0.695) as a built-in sanity check. The absolute numbers drop uniformly under the strict rule (both digits must land), but the **system ranking is invariant** to the jersey rule (E3 $>$ E1b $>$ Gemini-3.1-pro $>$ E1 $>$ Gemini-2.5-pro under both): our conclusions do not depend on the lenience of the matcher.

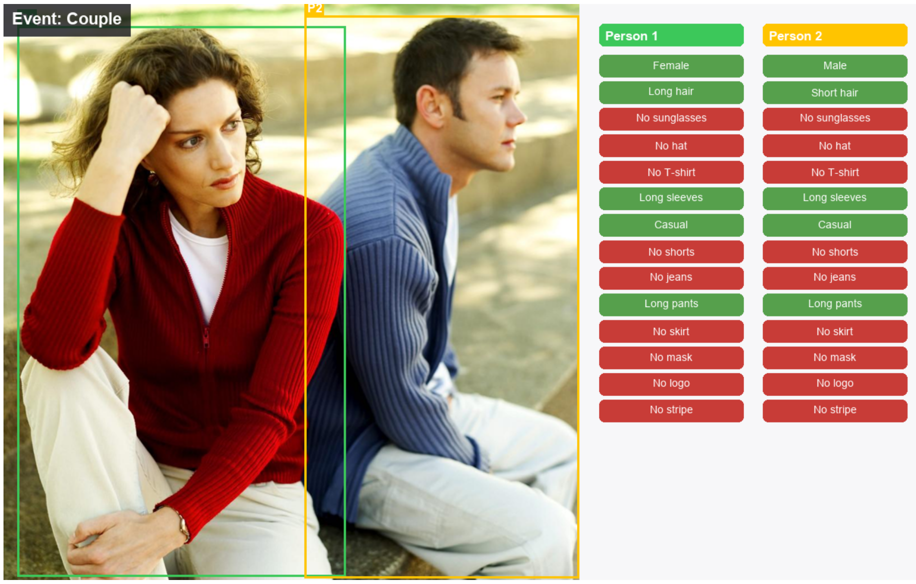

Fig. S10: Per-person attribute prediction on WIDER-Attribute (green = predicted true, red = predicted false; the two-valued attributes—gender, hair, sleeve, dress code—show the predicted pole and are drawn green). All boxes and attributes are model predictions from the frozen WIDER checkpoint (end-to-end, no ground-truth boxes). Each attribute set is emitted inside its person's block of the same grammar-constrained sequence, so attribute–person binding is native, exactly as on sports.

5 Baseline Prompt and Retry Policy

For full reproducibility, the verbatim prompt sent to the Gemini APIs (prompt version `eccv_v1`) is reproduced below. Decoding uses temperature 0; on a JSON-parse failure the call is retried up to 3 times with the temperature stepped by +0.3 per attempt. On the medium Gemini-3.1 configuration this retry path was triggered on 10.1% of frames (57/567; 66 extra calls, +11.6% call overhead) and left 0 frames unparsed after retries. Only the JSON is scored; any prose is discarded. Throughout the paper we quote the medium-resolution Gemini-3.1-pro configuration (its strongest); the thinking-disabled variant is weaker on detection (det-F1 0.65 / AP@50 0.45). The released caches record the exact API model identifiers, media-resolution settings, and run timestamps (July 2026), pinning the version of each API baseline.

First-attempt reliability and invalid geometry (full accounting). The thinking-disabled Gemini-3.1 variant returns *empty* output on 8.5% of frames (48/567); the medium configuration used throughout is never empty but required at least one JSON-repair retry on 10.1% of frames (above). The Gemini-2.5-pro run predates the parse-retry instrumentation (its log records network-level retries only), so its first-attempt rate is unrecorded—the - in Table 1 of the main paper.

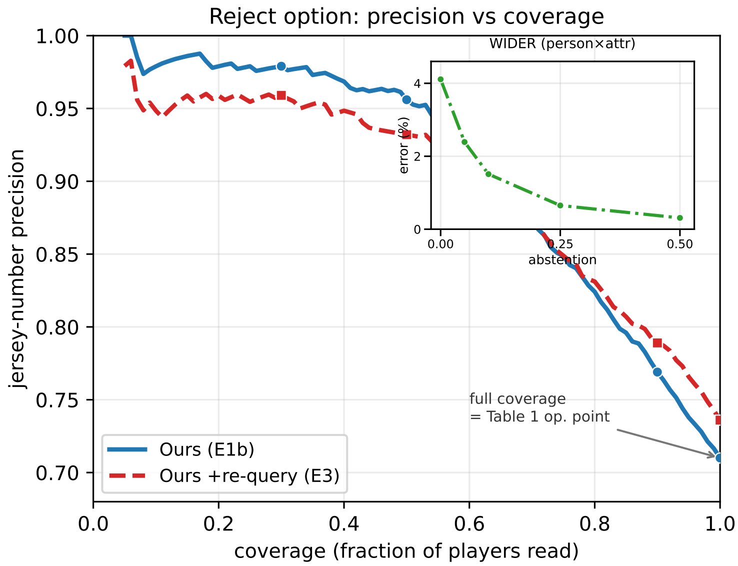

Fig. S11: Reject option, referenced from §5.4 of the main paper: jersey precision vs. coverage (rank by *jersey_conf*, abstain on the least-confident; precision is over scored digit reads, not the any-OCR precision of Table 1 of the main paper—definitions in Sec. 7, operating points in Tab. S6). E1b solid, re-query E3 dashed; inset: same mechanism on WIDER.

Degenerate (zero-area) boxes: Gemini-3.1 emits at least one on 22.2% of frames and *all* of a frame's boxes are degenerate on 8.1%; the rate for Gemini-2.5-pro is 0.5% of predicted players. Both oracle-tier cascades return a non-empty, well-formed parse on every frame (their fragility is association, not well-formedness).

Table S7: Confidence-aggregation ablation (reviewer-requested): the same unmasked-replay per-token probabilities aggregated by geometric mean (Eq. (1) of the main paper), arithmetic mean, and min-probability, scored over the same 952 predicted digit jerseys (fresh decode of the E1b checkpoint on the frozen test; the geometric-mean column reproduces the cached operating points of the main paper within ± 0.004). All three aggregators coincide to three decimals: predicted jersey fields span one–two BPE tokens, and on single-token fields every aggregator is identical by definition—the choice is immaterial in this regime, and Eq. (1)'s geometric mean is retained as the general length-normalized form (it matters only for longer OCR fields).

Aggregator	AUROC	ECE	P@1.0	P@0.9	P@0.7	P@0.5
Geometric mean (ours)	0.875	0.123	0.713	0.771	0.875	0.954
Arithmetic mean	0.875	0.123	0.713	0.771	0.875	0.954
Min-probability	0.875	0.123	0.713	0.771	0.875	0.954

Table S8: Jersey precision at matched coverage. At the recall each Gemini config operates at, our reject option (rank by *jersey_conf*, abstain on the least-confident) attains higher precision; the APIs expose no per-read confidence and cannot move off their single operating point. Any-OCR digit rule, same matcher, shared GT denominator ($N = 925$: the ground-truth players in the frozen test carrying a legible digit jersey after '?'-cleaning, of the 1,266 total); from cached reads, no re-inference. Full-coverage points reproduce Table 1 of the main paper.

System	jersey recall	jersey precision
Gemini-2.5-pro	0.612	0.829
Ours E1b (matched)	0.612	**0.920**
Ours E3 (matched)	0.612	0.913
Gemini-3.1-pro	0.645	0.860
Ours E1b (matched)	0.645	**0.906**
Ours E3 (matched)	0.645	0.899

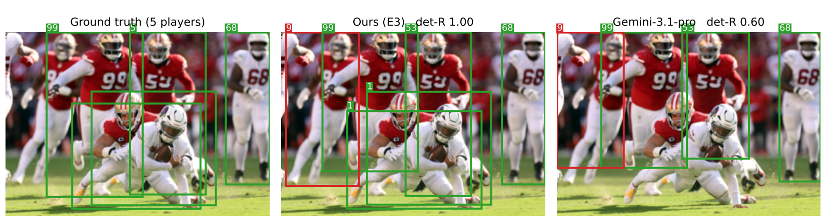

Fig. S12: Qualitative comparison on a crowded frame, referenced from §5.1 of the main paper (ground truth | ours with crop re-query | Gemini-3.1-pro). *Box colours:* green = prediction matched to a ground-truth player, red = unmatched prediction; box tags are the read jersey numbers. On this frame our detector localizes and reads the small, occluded athletes the API drops (detection recall 1.0 vs. 0.6); the size effect is analyzed in §5.2 of the main paper.

Table S9: Cascade reading quality by player-size quartile (`full_frame` oracle tier; jersey-F1 / OCR-text-F1). Detection recall is GT in every bucket. Buckets are GT-box-area quartiles (*tiny*/*small*/*med*/*large*) split at normalized-area edges 0.122, 0.232, 0.375 ($\approx$ 317 players each). Source: the guarded cascade run (Sec. 10).

System	Det-R	tiny	small	med	large
Florence-2 cascade (`florence_ff`)	GT	0.430/0.275	0.533/0.336	0.532/0.333	0.566/0.398
Specialist cascade (`paddle_ff`, PP-OCRv5)	GT	0.309/0.241	0.431/0.251	0.514/0.266	0.539/0.333

Table S10: Jersey-digit AER decomposition (frozen test, 4046-item GT vintage; the guarded cascade run, Sec. 10). The all-text AER splits into digit-swap + digit-unmatched + team-bleed (reconciliation identity, verified per system). Headline = digit-AER; team-bleed excluded. *Correct digit reads* counts emitted OCR items, not players, so it exceeds the 925 ground-truth players carrying a legible jersey (Tab. S8) for the systems that read more. The oracle-box cascades' digit-unmatched is 0 *as measured* (not by construction): we count, from the pre-stitch reads, every correct-digit read whose quad overlaps a player body that the IoSmaller stitch dropped—the cascade analogue of the APIs' grounded-to-nothing digit—and the count is 0 for both because on-body digits clear the 0.5 threshold (only off-body scoreboard/crowd text is dropped). So their digit-AER is matched-swaps-only as measured, yet still does not beat E1b (0.057). A bar-chart view is omitted because the oracle cascades' AER is recall-conditioned (they read only 32–37% of numbers), so it cannot be shown as a comparable bar.

	end-to-end					oracle cascade	
Quantity	E1	E1b	E3	Gem-2.5	Gem-3.1	Flor.	Padd.
all-text AER	0.099	0.085	0.097	0.249	0.202	0.070	0.057
all misassoc (n)	287	249	303	624	454	80	48
digit-swap	63	45	50	21	6	23	25
digit-unmatched	10	14	24	197	177	0	0
team-bleed (excl.)	214	190	229	406	271	57	23
correct digit reads	985	1030	1087	901	857	384	333
digit-AER	0.074	**0.057**	0.068	0.242	0.214	0.060	0.075
matched-only (not head.)	0.064	0.044	0.046	0.023	0.007	0.060	0.075

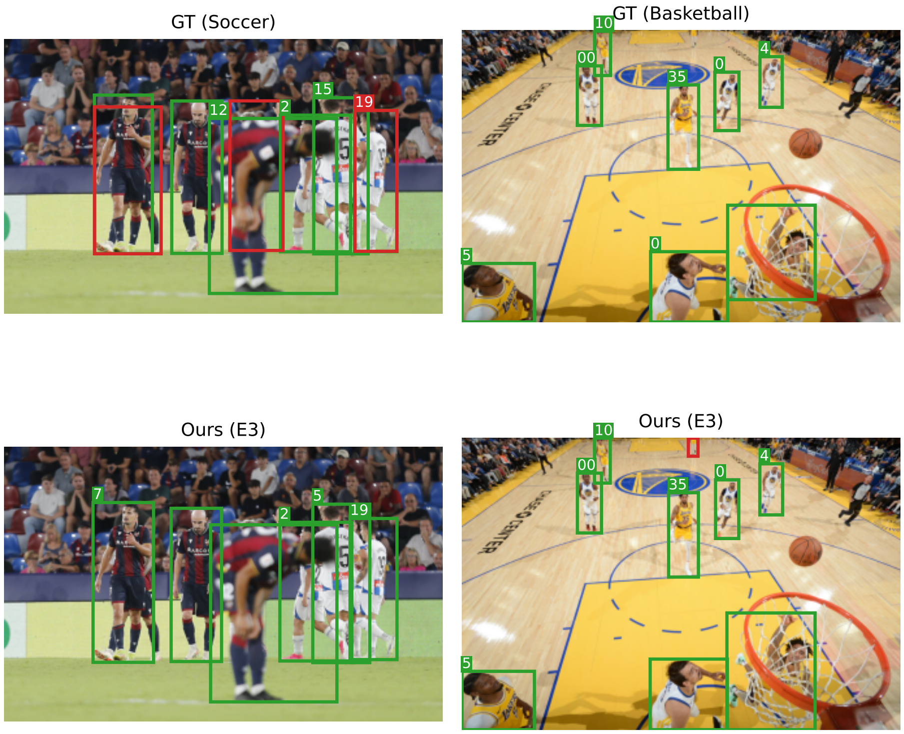

Fig. S13: Representative residual failures (ground truth, top; ours, bottom). *Box colours:* green = prediction matched to a ground-truth player, red = unmatched prediction; box tags are the read jersey numbers. **Left:** heavy occlusion (GT overlap ≈ 0.98) where a correctly-read number is bound to the wrong athlete—the residual perception AER of the main paper. **Right:** tiny jerseys that remain illegible even after the crop re-query, where the reads truncate (“12”→“7”, “15”→“5”). These are the honest boundary cases behind the small-player deficit, not systematic cascade-style identity swaps.

Table S11: Conditioning of residual misassociated *jersey digits* (frozen test, 4046-item GT). "matched-swap" = on a matched but wrong player; "unmatched" = grounded to no valid box. IoSmaller (IoS) = intersection/ min(area); centre distance and boxes are normalized; crowding = # ground-truth boxes overlapping the emitter. Source: the guarded cascade run (Sec. 10).

Condition	E1b	E3	Gem-2.5	Gem-3.1
Digit misassociations (n)	59	74	218	183
Matched-swap (%)	76.3	67.6	9.6	3.3
Unmatched / no box (%)	23.7	32.4	90.4	96.7
Unidentifiable, GT–GT IoS>0.5 (%)	15.3	37.8	6.9	1.6
Resolvable (%)	84.7	62.2	93.1	98.4
Owner-overlap > 0 (%)	62.7	86.5	94.5	7.1
Non-overlapping (%)	37.3	13.5	5.5	92.9
Median IoS (overlapping)	0.77	0.88	0.75	0.71
Median centre dist.	0.24	0.13	0.12	0.55
Median crowding	2	2	2	0

Table S12: Jersey precision/recall/F1 under the strict exact-digit rule and the any-OCR digit-token rule (frozen test, 4046-item GT). Same IoU matcher for every system; the ranking is identical under both rules. Gemini-3.1-pro is the medium-resolution configuration.

	strict exact-digit			any-OCR		
System	P	R	F1	P	R	F1
Ours E1 (+quads)	0.662	0.667	0.665	0.717	0.721	0.719
Ours E1b	0.692	0.697	0.695	0.755	0.759	0.757
Ours E3 (+re-query)	0.727	0.739	**0.733**	0.786	0.799	**0.792**
Gemini-2.5-pro	0.759	0.538	0.630	0.829	0.612	0.704
Gemini-3.1-pro	0.787	0.588	0.673	0.860	0.645	0.737

```
You are analyzing ONE sports photograph. Return ONLY a valid JSON object,
no prose, no markdown fences.

CRITICAL ANTI-HALLUCINATION RULES:
- Do NOT use outside knowledge. Do NOT guess player names, team names,
  leagues, game context, or famous kits.
- Only output what is directly visible in the image. If a value is not
  clearly readable/visible, set it to null (or [] for lists). Never invent.

OUTPUT JSON SCHEMA:
{
  "sport_type": "Basketball | Soccer | Hockey | American Football |
                Baseball | Tennis | Rugby | Cricket | Volleyball | Other",
  "scene_type": "In-Game | Warm-ups | Player Arrivals | Press Conference |
                Interview | Locker Room | Winning Ceremony | Other",
  "description": "<one concise sentence; no real names/teams>",
  "players": [
    {
      "player_index": <1-based index>,
      "team_affiliation": "Team A | Team B | Other",
      "bbox": {"x": <left 0-1>, "y": <top 0-1>, "w": <0-1>, "h": <0-1>},
      "jersey_facing": "Front | Back | Side",
      "jersey_number": "<clearly legible digits only, else null>",
      "ocr_items": [ {"text": "<readable text ON this player>"} ]
    }
  ]
}

RULES:
- DETECTION SCOPE: detect ATHLETES only; exclude referees, officials,
  coaches, staff, and off-field people (crowd/fans, bench staff).
- bbox: tight box around the whole visible body, normalized 0-1, origin
  top-left; x,y are the top-left corner, w,h are width/height.
- team_affiliation: cluster the competing athletes into "Team A"/"Team B"
  strictly by kit/jersey-color similarity; use "Other" ONLY when the team
  is genuinely unclear. Be consistent. Do NOT map real-world team names.
- jersey_number: ONLY if clearly legible; digits only; else null.
- ocr_items: EVERY clearly readable text fragment physically on the player
  (include partial). Never invent text; empty list if nothing readable.
- Output strictly the JSON object. No commentary, no markdown.
```

6 Dataset Composition

Table S13 gives per-sport frame and player counts by split, under the canonical six-way sport bucketing (gridiron and American football merged into *football*; ice hockey merged into *hockey*; sparse residual sports folded into *other*). In-game footage dominates the corpus (≈86% of frames).

7 Metric Definitions in Full

This section expands the metric summary of §4 of the main paper; the association error rate is defined in the main text and decomposed in Sec. 3.1.

Detection AP@50 (for completeness). The main comparison (§4 of the main paper) scores detection by precision/recall/F1; we collect AP@50 for the learned systems here (Tab. S14). The re-query variant leaves detection unchanged, and the oracle-boxed cascades are handed ground-truth boxes, so AP@50 is not defined for them.

Table S13: Per-sport counts by split, reported as *frames / players*, under the canonical six-way sport bucketing. Frame counts match the E0 dataset report; the corpus totals 5,728 frames and 12,999 players.

Sport	Train	Val	Test	Total
Hockey	1,225 / 2,696	153 / 328	151 / 325	1,529 / 3,349
Soccer	1,152 / 3,032	145 / 405	143 / 377	1,440 / 3,814
Football	1,147 / 2,430	144 / 301	143 / 282	1,434 / 3,013
Tennis	521 / 639	66 / 80	65 / 86	652 / 805
Basketball	508 / 1,560	64 / 202	63 / 194	635 / 1,956
Other	31 / 51	5 / 9	2 / 2	38 / 62
Total	4,584 / 10,408	577 / 1,325	567 / 1,266	5,728 / 12,999

Table S14: Detection AP@50 on the frozen 567-frame test set. AP@50 is VOC 11-point AP computed with uniform prediction scores (generative systems and the API JSON emit no ranking score); we report it for completeness—F1 is the operative detection metric (Table 1 of the main paper). †Oracle GT boxes—AP@50 undefined.

	Ours (E1b)	Gemini-2.5-pro	Gemini-3.1-pro	Cascades
AP@50	**0.886**	0.480	0.528	GT†

Team purity (permutation invariance). Predicted team flags are cluster labels: which visual cluster is called "A" is arbitrary, so a systematically flipped labeling is not a semantic error. We therefore score purity *per frame*: over the frames with at least two ground-truth teams, we build the confusion matrix between predicted and ground-truth team labels on the IoU-matched players and take the best label correspondence by Hungarian assignment (maximizing the matched trace); the frame's purity is its matched count over its matched players, and we micro-average over all matched players across those frames. Exact team accuracy uses the raw labels with no correspondence; purity $\geq$ accuracy by construction, and the gap measures how much of the apparent team error is a per-frame label permutation rather than a genuine misclustering. Concretely: exact accuracy (Table 1 of the main paper: 0.809) uses raw labels over all 1203 matched players; purity (0.952) applies the per-frame correspondence over the 687 matched players in the 226 frames with at least two ground-truth teams; and the margin analysis's aligned accuracy (0.970, Tab. S2) applies the same correspondence over all 1203 matched players (single-team frames included, which align trivially)—so the aligned figure exceeds purity by the contribution of those trivially-aligned single-team frames, not a different alignment.

OCR-text matching. Predicted and ground-truth on-body strings are pooled per frame and matched by Hungarian assignment that maximizes pairwise normalized-Levenshtein similarity; a pair is accepted as a true positive only if its similarity clears $\tau = 0.7$, and unmatched predictions/ground-truths become false positives/negatives. Precision, recall, and F1 are micro-averaged over all pooled items across the 567 frames, so frames with more text contribute proportionally.

Scene description. The free-form scene description is scored by SBERT cosine similarity (`all-MiniLM-L6-v2`) between predicted and ground-truth descriptions; our headline model scores 0.808, stable across all fine-tuned arms (0.806–0.808). It is deliberately excluded from the pre-registered *val_score* and from the main-paper comparison table (Table 1); scene metrics are reported here rather than in the headline table.

Confidence metrics. For the reject-option and calibration analysis (§5.4 of the main paper) we report, on the scored jersey reads: (i) *AUROC* of *jersey_conf* as a correct-vs-incorrect discriminator; (ii) *expected calibration error* with 15 equal-width bins between mean confidence and empirical accuracy; and (iii) *precision at fixed coverage*—ranking players by *jersey_conf*, abstaining on the least-confident fraction, and reporting jersey precision on the retained reads. *Scored reads* = IoU-matched players with a primary digit read (952 for E1b, 970 for E3); this is the precision base of the reject curve (Tab. S6) and of Fig. S5(b), and is distinct from Table 1 of the main paper's any-OCR precision denominator (0.710 vs. 0.755 at full coverage is this base difference, not a discrepancy). The structural signals *loc_conf* and the objectness margin are scored by the same AUROC protocol as negative results.

8 Bootstrap Confidence Intervals

We report the paired frame-level bootstrap behind the statistical-reliability paragraph of §5 in full (Tab. S15). We draw 10,000 resamples of the 567 frozen-test frames with replacement; each resample scores *every* system on the same frame multiset (paired), and we take the 2.5/97.5 percentiles of the per-resample delta. Deltas are signed so a positive value always means "ours (E1b) better": for the three F1 metrics $\Delta = \text{ours} - \text{baseline}$; for the association error rate (lower is better) $\Delta = \text{baseline} - \text{ours}$. The computation is offline from the cached per-frame predictions using the same matcher and scorers as the main comparison (§5.1 of the main paper), and a gate asserts the full-sample point estimates reproduce the published numbers before any resampling.

9 Implementation Details

All self-hosted models are the same Florence-2-large (0.77B) backbone with frozen vision and text towers. Inference uses greedy decoding under the finite-state grammar; accuracy metrics are computed in `fp32`, while the reported latency is a separate lightweight `fp16` on-GPU decode, both on a single NVIDIA A100-80GB (PCIe). Mean per-frame on-GPU latency (`fp16`) is 1224 ms for our headline single-pass model (E1b) and 1274 ms with the crop re-query (E3, at

Table S15: Paired bootstrap 95% CIs of the headline deltas (Δ = Ours (E1b) − baseline; positive favours us). Detection is not compared against the cascades, which are handed ground-truth boxes (†). Every headline interval excludes 0 except two—jersey F1 vs. Gemini-3.1-pro (medium-resolution) and digit-AER vs. the oracle cascades—both reported as such in the main text. ‡AER delta = baseline − ours.

Δ (95% CI)	Gemini-2.5	Gemini-3.1-pro	Specialist	Florence-2
Detection F1	+0.30 [+0.27, +0.33]	+0.22 [+0.18, +0.25]	GT†	GT†
Jersey F1	+0.05 [+0.02, +0.09]	+0.02 [−0.02, +0.06]	+0.30 [+0.26, +0.34]	+0.24 [+0.20, +0.27]
OCR-text F1	+0.11 [+0.08, +0.13]	+0.09 [+0.06, +0.11]	+0.30 [+0.28, +0.33]	+0.24 [+0.22, +0.26]
AER (digits)‡	+0.18 [+0.15, +0.22]	+0.16 [+0.11, +0.20]	+0.02 [−0.02, +0.05]	+0.00 [−0.03, +0.03]

1.93 passes/frame); the quad-emitting variant (E1) is slower (1710 ms). These `fp16` figures are the headline latencies (the main comparison, §5.1 of the main paper); the `fp32` generation of the same E1b model is 1273 ms (the base for the confidence-replay share below and the grammar-overhead accounting in the ablations, §5.3 of the main paper)—the two are one model at two decode precisions, not a discrepancy. The Gemini rows are API wall-clock (network + endpoint) at temperature 0: 26.0 s (2.5-pro) and 8.4 s (3.1-pro, medium-resolution). For the oracle cascades, reading+team+stitch runs in 89 ms (PaddleOCR PP-OCRv5) and 708 ms (Florence-2 <OCR_WITH_REGION>) per frame in `fp32`, with detection excluded (oracle boxes), so these are not comparable with the end-to-end rows. The Gemini token budget for the frozen test was 0.61M input + 0.23M output tokens, i.e. $7.11/1k frames at list price for the medium-resolution 3.1 configuration.

Model architecture. Florence-2-large couples a DaViT vision encoder to a BART-style encoder–decoder (Fig. S14). Frames are processed at 768 × 768 and tokenized to 577 visual embeddings; these are concatenated with the embedded task prompt and fused by the (frozen) transformer encoder, after which the decoder autoregressively emits our serialization. The prompt convention is: the task prompt (<SPORTS_UNIFIED> "Describe the sports scene . . . ") is *encoder* text input, while the *decoder* target begins at <sport>—a useful sanity signature is that epoch-1 predictions start </s><s><sport>. We freeze the vision tower and the BART encoder and fine-tune only the decoder and the shared token embeddings (Tab. S18); the schema caps at 12 players, ≤ 17 OCR items per player, and ≤ 96 scene-description tokens. Player blocks are serialized largest-first (bbox area, descending) and OCR items largest-quad-first; the replay's softmax is *unmasked* (full vocabulary), while the team and objectness margins are renormalized over their legal token sets. The token-embedding matrix is tied across the BART encoder and decoder (and the output projection), so fine-tuning it updates the single shared table used on both sides—this is why the frozen encoder still sees the warm-started structural-token rows. (The decoder's trainable parameter count was not recorded in the run metadata and is therefore omitted.)

Table S18 lists the fine-tuning recipe for our models. It is identical across E1/E1b (the pair differs by a single flag: quad supervision on/off and its grammar

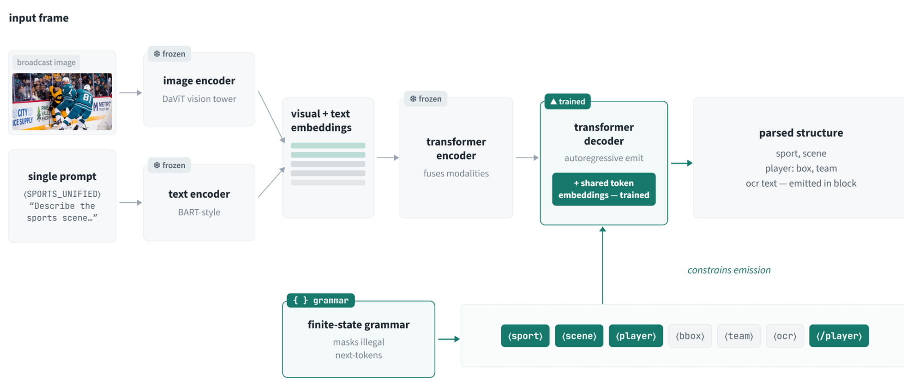

Fig. S14: System overview. Frozen DaViT vision encoder and frozen text encoder feed the transformer encoder (also frozen); only the decoder and the shared token embeddings are fine-tuned, so the pretrained visual priors are preserved and training teaches the decoder the output schema—"vocabulary alignment" rather than representation learning. At decode time the finite-state grammar masks illegal continuations, guaranteeing the serialization of §3.1.

branch; the on/off comparison is Tab. S5, the full quad ablation referenced from §5.3 of the main paper) and is the same recipe transplanted to WIDER-Attribute; the LoRA arms (Tab. S4) freeze this base and train only the injected adapters. The selection aggregate *val_score* is the unweighted mean of five task metrics: detection F1, team accuracy, jersey-number F1, multi-OCR text F1, and scene accuracy.

Measured confidence-replay cost. The per-field confidence (§3.2, §5.4 of the main paper) is computed by one teacher-forced forward over the generated tokens, and it is genuinely cheap: on the frozen test the replay is ≈ 150 ms/frame (149.7 ms E1b, 152.3 ms E1, constant across the E1/E1b/LoRA arms), which is 11.8% of E1b `fp32` generation (1273 ms) and 8.4% of E1 generation (1808 ms). It is a *fixed* cost because the replay scores all token positions in a single parallel forward, whereas generation is autoregressive and grows with sequence length; E1b shows the highest replay *share* precisely because its no-quad generation is the fastest—the replay is the constant, generation length is the variable.

Inherited decode defaults (a practical warning). A practical warning for Florence-2 fine-tuners: the language model inherits `no_repeat_ngram_size`=3 and `early_stopping`=True from its text config, and `generate()` calls that pass only `max_new_tokens` keep them. A trigram ban makes the second instance of any repeated schema fragment (`</person><person><bbox>`. . .) illegal, silently capping multi-instance structured output at 1–2 entities regardless of training—in our early WIDER runs this pinned the task metric for 24 epochs. Set `no_repeat_ngram_size`=0 and `early_stopping`=False in every generate call, persist a cleaned `generation_config` with the checkpoint, and re-apply on load.

The start-token loop (the same lesson, most dramatic exhibit). Removing the finite-state grammar entirely is a more fundamental failure than the inherited-defaults trap above: even with those defaults corrected (`no_repeat_ngram_size`=0, `early_stopping`=False), free greedy decoding collapses: the model never emits a content token, producing an unbounded run of start/end markers until the 1024-token cap—e.g.

```
</s><s><s><s><s>...<s>
```

so every frame yields an empty parse, zero players, and $\approx 7\times$ the constrained latency (main paper's no-grammar ablation, §5.3). The grammar's first-token mask forbids exactly this continuation, which is why constrained decoding both enforces the schema *and* terminates generation—a failure the inherited-defaults fix alone does not prevent, and which the grammar removes by construction. The same collapse holds on WIDER-Attribute, and no grammar-free decode configuration averts it (Tab. S17). On the final WIDER checkpoint (30 test images, GT 8.3 persons/image), the grammar yields schema-valid output on every image (7.7 persons/image, 107 attributes); without it, greedy and beam-search (`num_beams`= 3) both fall into the identical start-token loop (zero persons, schema-valid 0), and beam search with the inherited `no_repeat_ngram_size`=3 breaks the loop only to degrade into malformed output (2.6 persons/image, schema-valid 0 on every image). The grammar is the sole configuration that produces schema-valid output; the others differ only in *how* they fail. Table S16 gives the full sports-domain table (demoted from the main paper's §5.3 for the camera-ready page budget; the prose there carries the headline numbers).

10 Reproducibility

Release scope. Release scope (what is and is not released, and why) is stated in §4 of the main paper; this section pins the operational provenance behind it.

All cascade and oracle baselines were produced by a single guarded run (four configurations: two cascades, two Geminis) covering all 567 test frames with zero errors; the frozen-test fingerprint matched, and reading/team/stitch latencies and the confidence replay are `fp32`. Ours (E1/E1b/E3) are scored offline from their own caches—seven systems total pass through a reconciliation gate: before any analysis, a per-record reconstruction must reproduce the published digit-AER and the three decomposition counts (digit-swap, digit-unmatched, team-bleed) exactly, the reconciliation identity must hold, and the all-text AER must still match; a drift on either definition aborts the run. A second gate asserts the cascades' measured stitch-dropped on-body digit count (Tab. S10; 0 for both),

Table S16: Sports no-grammar ablation (E1b, same greedy config; the only change is the finite-state constraint). Unconstrained decoding emits an unbounded start-token loop: every output is empty (schema validity is vacuous—an empty output parses), zero players, 7× the latency; AER and degenerate-box rate are undefined (no reads). The constrained column scores the `fp32` frozen-test decode; its jersey F1 differs by ≤ 0.001 from Table 1 of the main paper (a separate `fp32` frozen-test decode of the same checkpoint).

Metric	Constrained	No grammar
Schema-valid rate↑	1.000	–
Empty / no-parse rate↓	0.000	1.000
Persons per image	2.23	0.00
Detection F1↑	0.950	0.000
Jersey F1 (any-OCR)↑	0.756	0.000
Jersey F1 (strict)↑	0.694	0.000
Jersey-digit AER↓	0.057	–
Degenerate-box rate↓	0.000	–
Latency (ms/frame, fp32)	1273	8854

Table S17: WIDER-Attribute grammar on/off on the final checkpoint (30 test images, GT 8.3 persons/image). *persons* = mean predicted persons per image; *valid* = schema-valid rate. Only the grammar yields schema-valid output; the three grammar-free decode configurations all fail at schema validity, differing only in failure mode.

Decode configuration	persons	valid	Failure mode
Grammar (greedy)	7.7	**1.00**	—
No grammar (greedy, `ngram`=0)	0.0	0.00	Start-token loop
No grammar (beam=3, `ngram`=0)	0.0	0.00	Start-token loop
No grammar (beam=3, `ngram`=3)	2.6	0.00	Malformed output

so a change that begins erasing on-body reads is caught rather than silently discounted. The per-crop re-assignment and the misassociation-conditioning and jersey-rule analyses (Secs. 3, 3.4 and 4) are offline recomputations from the cached predictions, each validated against the published numbers before use; the released caches and scripts reproduce them.

Table S18: Fine-tuning recipe (self-hosted models). The vision tower (DaViT) and the BART *encoder* are frozen; the BART *decoder* and the shared embeddings (including the warm-started structural tokens) are trained. Source: run meta of the E1/E1b runs.

Setting	Value
Backbone	Florence-2-large (0.77B)
Frozen	DaViT vision tower + BART encoder
Trained	BART decoder (full FT) + token embeddings
Token init	warm-start (mean-embedding) for new tokens
Optimizer	AdamW, lr 5×10^{-5}, weight decay 0.1
Schedule	cosine, 5% linear warm-up
Batch size	6
Epochs	20 (early stop on val loss, patience 3)
Selection	`best_model_score` (val aggregate)
Precision	`fp32` training/accuracy; `fp16` latency
Seed	42
Hardware	single NVIDIA A100-80GB